

1. Functional Requirements

The functional requirements define the core system services that SafeSight must provide in order to support industrial PPE compliance monitoring, hazard detection, incident validation, corrective action tracking, and reporting activities within monitored industrial environments.

Each requirement below is written as a testable system obligation and aligned with the scope established in D1.

FR-01

The system shall process live video streams received from connected industrial cameras in real time.

FR-02

The system shall detect PPE elements including helmet, vest, gloves, and goggles using the YOLO11 object detection model.

FR-03

The system shall detect hazard objects including fire, smoke, and forklift presence within monitored industrial zones.

FR-04

The system shall generate an alert when a PPE violation or hazard condition exceeds configured detection thresholds.

FR-05

The system shall automatically create an incident record for each generated alert including timestamp, monitored zone, detection type, and initial severity level.

FR-06

The system shall allow a Safety Officer to validate an incident by confirming or rejecting the generated alert.

FR-07

The system shall allow a Supervisor to assign corrective actions for confirmed incidents.

FR-08

The system shall allow authorized users to update incident status until closure.

FR-09

The system shall generate weekly and monthly compliance reports based on recorded incidents and violation trends.

FR-10

The system shall enforce role-based access control for all authenticated users.

FR-11

The system shall store evidence snapshots for confirmed incidents.

FR-12

The system shall maintain audit logs for critical actions including validation, status updates, and configuration changes.

2. Use Cases

The following use cases describe the major interactions between system actors and SafeSight. These use cases reflect the operational workflow defined in the project scope and ensure behavioral traceability between functional requirements and domain concepts.

UC-01 Detect Safety Violation**Goal**

To detect PPE violations or hazardous conditions from live video streams.

Primary Actor

Camera System

Preconditions

The camera stream is active and connected.

Trigger

A new video frame enters the system.

Main Success Scenario

1. The camera sends a live frame to the system.
2. The system processes the frame using the YOLO11 model.

3. PPE elements and hazard objects are detected.
4. The system identifies whether a violation or hazard exists.
5. The detection result is passed to alert evaluation.

Alternate Flow

If image quality is insufficient, detection confidence may fall below threshold.

Postconditions

A valid detection event is produced for evaluation.

UC-02 Generate Alert**Goal**

To create a system alert after a valid violation or hazard detection.

Primary Actor

System

Preconditions

A valid detection event exists.

Trigger

Detection confidence exceeds configured threshold.

Main Success Scenario

1. The system checks detection confidence.
2. Severity is determined according to detection type.
3. An alert is generated.
4. Notification data is prepared.

Alternate Flow

If confidence remains below threshold, no alert is generated.

Postconditions

An alert is available for incident creation.

UC-03 Validate Incident**Goal**

To confirm or reject a generated alert.

Primary Actor

Safety Officer

Preconditions

An alert exists in the system.

Trigger

The Safety Officer opens the alert.

Main Success Scenario

1. The Safety Officer reviews evidence.
2. The alert is confirmed or rejected.
3. Validation notes are entered.
4. Incident status is updated.

Alternate Flow

If the alert is incorrect, it is marked as false alarm.

Postconditions

Incident status becomes confirmed or rejected.

UC-04 Assign Corrective Action

Goal

To assign a corrective response for a confirmed incident.

Primary Actor

Supervisor

Preconditions

The incident is confirmed.

Trigger

Supervisor reviews incident details.

Main Success Scenario

1. The Supervisor reviews the incident.
2. A corrective action is selected.

3. Responsibility is assigned.

Alternate Flow

If responsible personnel cannot be identified, escalation is initiated.

Postconditions

Corrective action is recorded.

UC-05 Update Incident Status**Goal**

To manage incident lifecycle progression.

Primary Actor

Supervisor

Preconditions

Incident exists and remains active.

Trigger

Corrective action progress changes.

Main Success Scenario

1. Supervisor opens incident record.
2. Status is updated.
3. Resolution note is saved.

Alternate Flow

If issue persists, incident remains open.

Postconditions

Updated incident state is stored.

UC-06 Generate Compliance Report**Goal**

To produce compliance summaries and KPI reports.

Primary Actor

Safety Officer

Preconditions

Historical incident data exists.

Trigger

Report request is initiated.

Main Success Scenario

1. Reporting period is selected.
2. System aggregates incidents.
3. KPI summary is generated.

Alternate Flow

If no data exists, an empty report is produced.

Postconditions

A compliance report becomes available.

UC-07 Handle False Alert**Goal**

To manage false positive alerts.

Primary Actor

Safety Officer

Preconditions

Alert exists.

Trigger

Officer rejects alert.

Main Success Scenario

1. Alert evidence is reviewed.
2. Alert is rejected.
3. Rejection reason is stored.

Alternate Flow

If uncertainty remains, further review is requested.

Postconditions

Alert is recorded as false alarm.

UC-08 Manage System Configuration

Goal

To maintain operational settings.

Primary Actor

System Administrator

Preconditions

Administrator is authenticated.

Trigger

Configuration update is requested.

Main Success Scenario

1. Administrator updates threshold or zone parameters.
2. New values are validated.
3. Configuration is saved.

Alternate Flow

Invalid values are rejected.

Postconditions

System configuration is updated.

3. Non-Functional Requirements

The following non-functional requirements define the expected operational qualities of SafeSight.

NFR-01 Security

The system shall require authentication before access to protected functions.

NFR-02 Authorization

The system shall restrict critical actions according to user role permissions.

NFR-03 Performance

The system shall generate alerts within 2 seconds under normal operating conditions.

NFR-04 Reliability

The system shall automatically attempt camera reconnection after stream interruption.

NFR-05 Maintainability

The system shall allow threshold updates without modifying application code.

NFR-06 Auditability

The system shall record audit logs for critical system actions.

4. Data Requirements

The system stores operational and safety-related data required for incident management and reporting.

Key Data Objects

Incident

- incident_id
- timestamp
- zone
- type
- severity
- status

Alert

- alert_id
- confidence
- detection_class

Evidence

- snapshot_path
- short_clip_reference

User

- user_id

- role
- credentials

Corrective Action

- action_id
- description
- due_date

Report

- reporting_period
- KPI values

Data Assumptions

Confirmed incident evidence must be retained according to internal policy.
Access to stored evidence must be restricted by role.

5. Acceptance Criteria / Test Conditions

The following acceptance criteria define testable operational outcomes.

AC-01

Given a live stream, when a worker appears without helmet, then an alert shall be generated.

AC-02

Given fire detection above threshold, when validation begins, then an incident shall be created.

AC-03

Given a confirmed incident, when a supervisor assigns action, then corrective action shall be stored.

AC-04

Given a false alert, when rejected by Safety Officer, then status shall become false alarm.

AC-05

Given monthly reporting request, when data exists, then KPI summary shall be generated.

AC-06

Given unauthorized access attempt, when restricted action is requested, then access shall be denied.

6. Domain Explanation

SafeSight operates in the domain of industrial safety monitoring and incident management.

The system continuously observes monitored industrial zones and interprets real-world safety conditions through computer vision detection. Detected events become alerts, alerts become incidents after validation, and confirmed incidents trigger corrective actions and reporting processes.

The domain therefore connects physical safety conditions, operational responsibility, and structured digital incident control.

7. Business Rules

BR-01

An incident can be confirmed only by a Safety Officer.

BR-02

A corrective action can be assigned only after incident confirmation.

BR-03

An incident cannot be closed before corrective action completion.

BR-04

Only System Administrators may change system thresholds.

8. UML Models

Figure 1. Use Case Diagram for SafeSight

The following use case diagram illustrates the primary actors interacting with SafeSight and the main operational functions provided by the system.

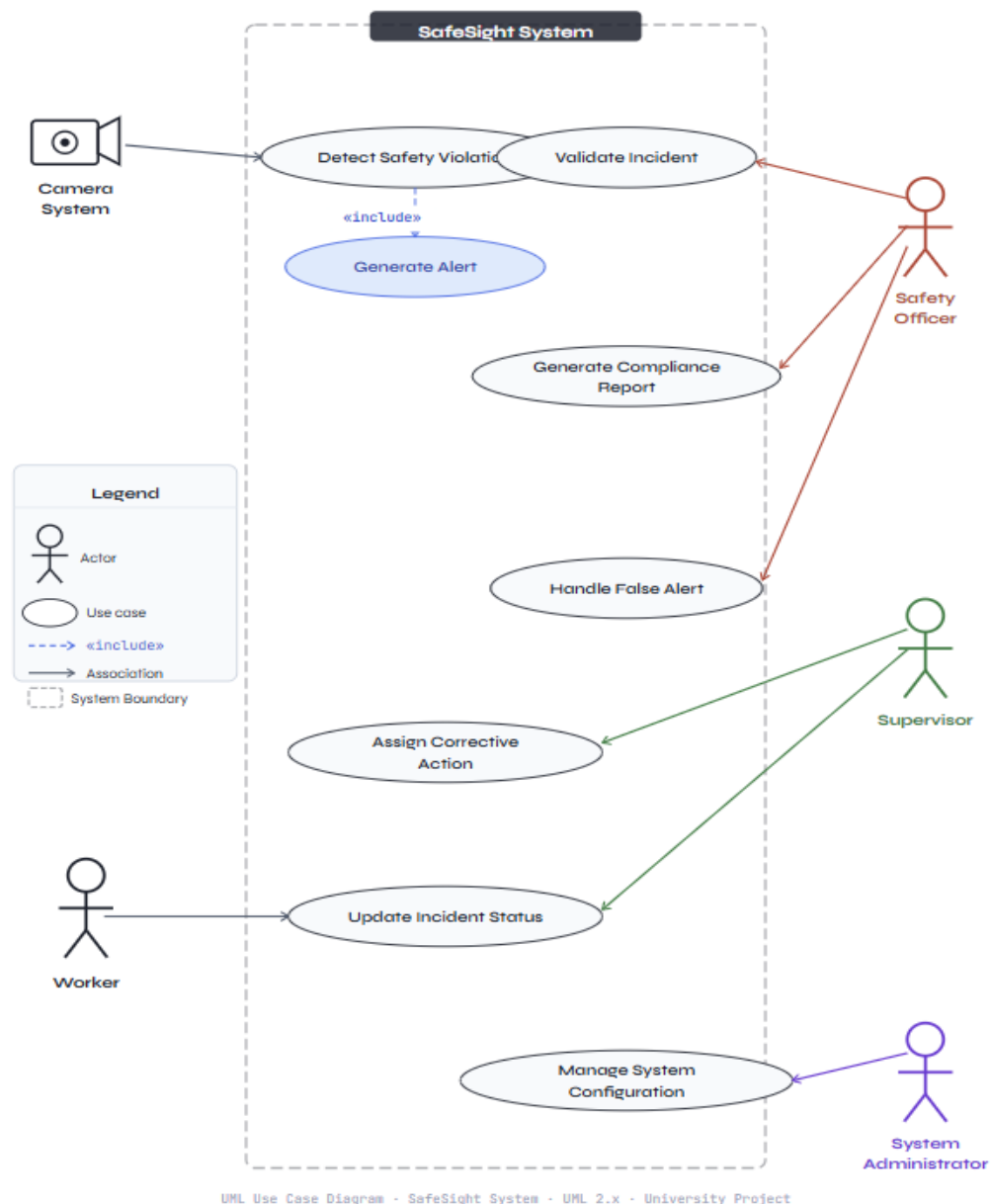


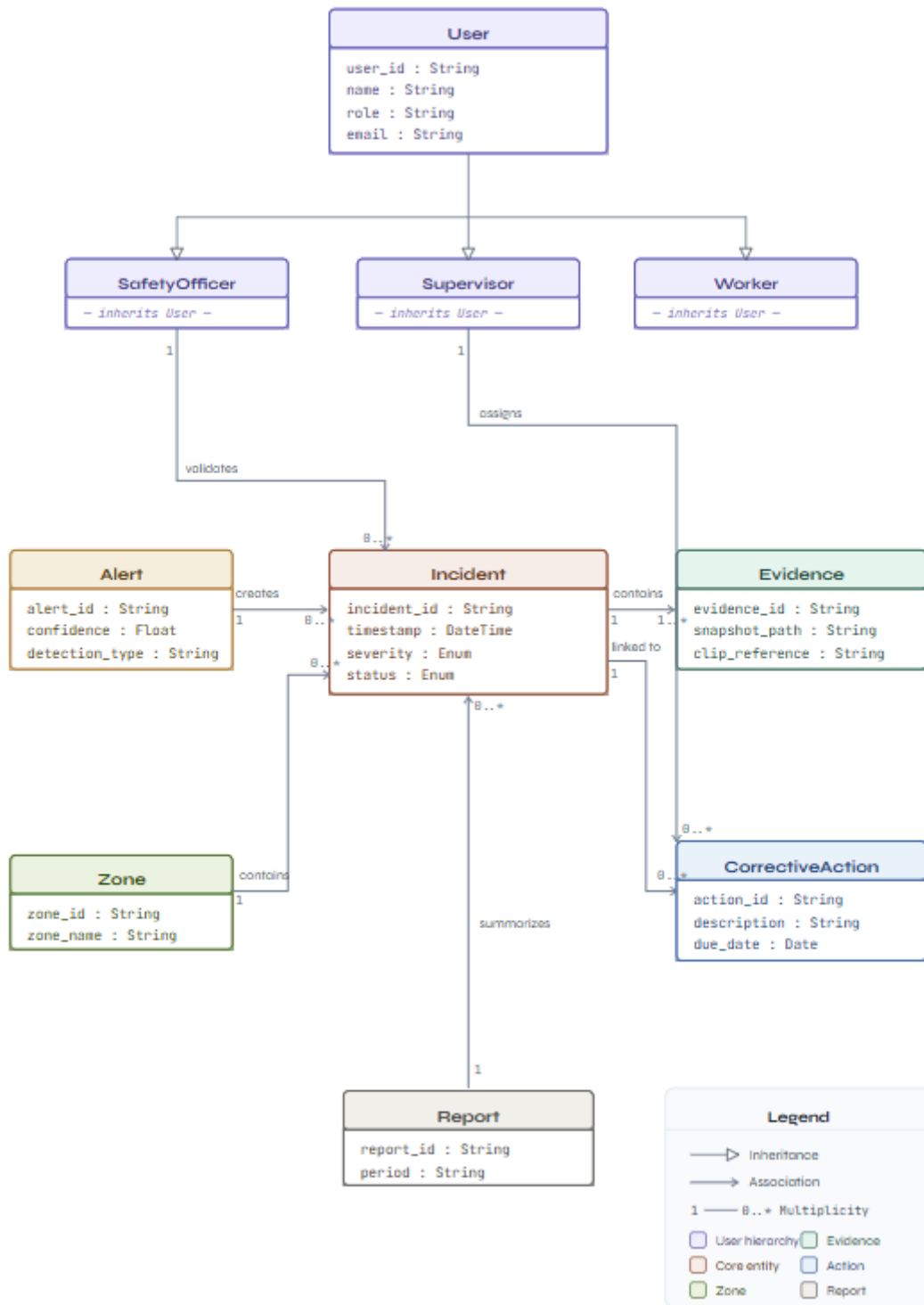
Figure 2. Activity Diagram for Incident Validation

The following activity diagram represents the incident validation workflow performed by the Safety Officer after a system-generated alert is created



Figure 3. Conceptual Class Diagram of SafeSight

The conceptual class diagram below represents the main domain entities, their relationships, and the conceptual structure of SafeSight.



9. Mapping Tables

Table 1. Functional Requirements to Use Case Mapping

Functional Requirement	Related Use Case(s)
FR-01 Process live video streams	UC-01 Detect Safety Violation
FR-02 Detect PPE elements	UC-01 Detect Safety Violation
FR-03 Detect hazard objects	UC-01 Detect Safety Violation
FR-04 Generate alerts	UC-02 Generate Alert
FR-05 Create incident record	UC-02 Generate Alert, UC-03 Validate Incident
FR-06 Validate incident	UC-03 Validate Incident
FR-07 Assign corrective action	UC-04 Assign Corrective Action
FR-08 Update incident status	UC-05 Update Incident Status
FR-09 Generate compliance reports	UC-06 Generate Compliance Report
FR-10 Enforce role-based access control	UC-08 Manage System Configuration
FR-11 Store evidence snapshots	UC-03 Validate Incident
FR-12 Maintain audit logs	UC-03 Validate Incident, UC-05 Update Incident Status, UC-08 Manage System Configuration

Table 2. Use Case to Domain Concept Mapping

Use Case	Related Domain Concepts
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UC-01 Detect Safety Violation	Alert, Zone
UC-02 Generate Alert	Alert, Incident, Evidence
UC-03 Validate Incident	Incident, SafetyOfficer, Evidence
UC-04 Assign Corrective Action	Incident, CorrectiveAction, Supervisor
UC-05 Update Incident Status	Incident, CorrectiveAction, Supervisor
UC-06 Generate Compliance Report	Report, Incident, Zone
UC-07 Handle False Alert	Alert, Incident, SafetyOfficer
UC-08 Manage System Configuration	User, Zone